caddis use sticks as building materials, others fragments of dead leaves, others small snail shells. But perhaps the most impressive caddis houses are the ones built in local stone. The caddis chooses its stones carefully, rejecting those that are too large or too small for the current gap in the wall, even rotating each stone until it achieves the snuggest fit.

Incidentally, why does this impress us so? If we forced ourselves to think in a detached way we surely ought to be more impressed by the architecture of the caddis's eye, or of its elbow joint, than by the comparatively modest architecture of its stone house. After all, the eye and the elbow joint are far more complicated and 'designed' than the house. Yet, perhaps because the eye and elbow joint develop in the same kind of way as our own eyes and elbows develop, a building process for which we, inside our mothers, claim no credit, we are illogically more impressed by the house.

Having digressed so far, I cannot resist going a little further. Impressed as we may be by the caddis house, we are nevertheless, paradoxically, less impressed than we would be by equivalent achievements in animals closer to ourselves. Just imagine the banner headlines if a marine biologist were to discover a species of dolphin that wove large, intricately meshed fishing nets, twenty dolphin-lengths in diameter! Yet we take a spider web for granted, as a nuisance in the house rather than as one of the wonders of the world. And think of the furore if Jane Goodall returned from Gombe stream with photographs of wild chimpanzees building their own houses, well roofed and insulated, of painstakingly selected stones neatly bonded and mortared! Yet caddis larvae, who do precisely that, command only passing interest. It is sometimes said, as though in defence of this double standard, that spiders and caddises achieve their feats of architecture by 'instinct'. But so what? In a way this makes them all the more impressive.

Let us get back to the main argument. The caddis house, nobody could doubt, is an adaptation, evolved by Darwinian selection. It must have been favoured by selection, in very much the same way as, say, the hard shell of lobsters was favoured. It is a protective covering for the body. As such it is of benefit to the whole organism and all its genes. But we have now taught ourselves to see benefits to the organism as incidental, as far as natural selection is concerned. The benefits that actually count are the benefits to those genes that give the shell its protective properties. In the case of the lobster this is the usual story. The